

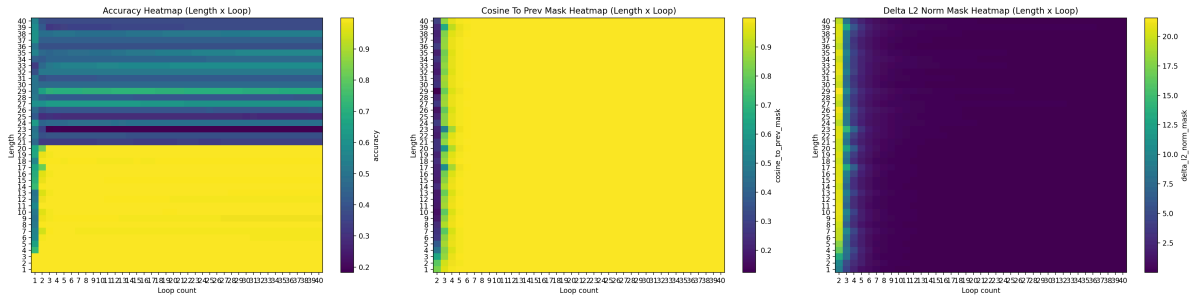
PonderNet

1. 更长的上限?
2. constant depth下能决定，那么长度会被精度和dim限制住
3. adaptive exit push模型converge，限制表达能力？好处是converge？会导致什么问题？
4. adaptive exit会harm extrapolation，会破坏到什么程度？
5. **parity**和**copy**一起训？
6. exit只是function of length & loop
7. **copy**变成指定输出第n个 **one-bit** 输出 → **.retrive**
8. 二进制乘法？
9. reachability
10. **decouple length & difficulty**
11. 什么时候用continuous的时候应该decode coconut, transformer，提升是否来自latent？不用hidden

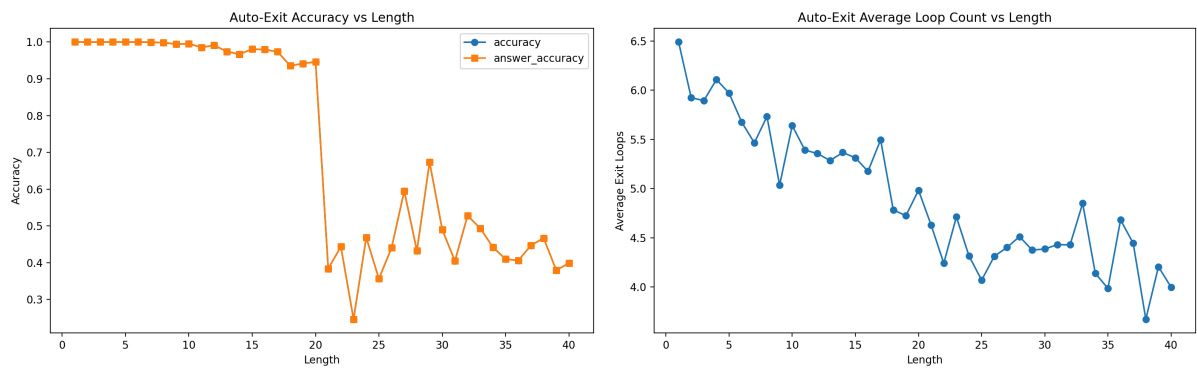
$$L = \underbrace{\sum_{n=1}^N p_n \mathcal{L}(y, \hat{y}_n)}_{L_{Rec}} + \beta \underbrace{KL(p_n || p_G(\lambda_p))}_{L_{Reg}}$$

PonderNet虽然是在RNN上实现的parity，但是效果感觉比较好，拿到我们这里也试试

如果是对mask进行mean pooling之后接Exit heat，训练得到的结果是

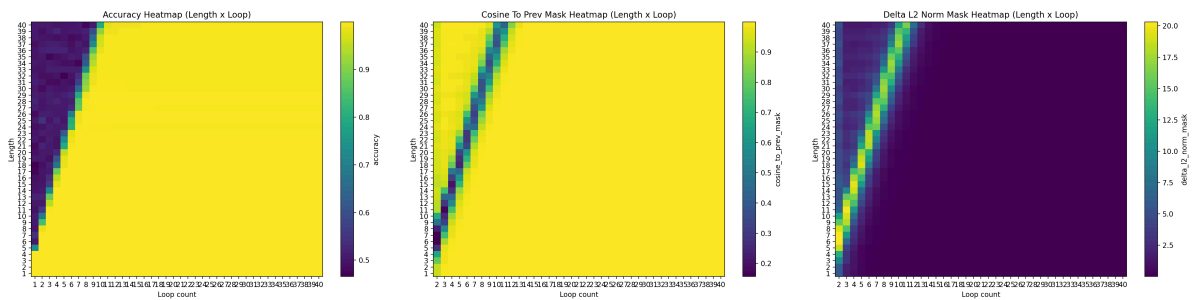


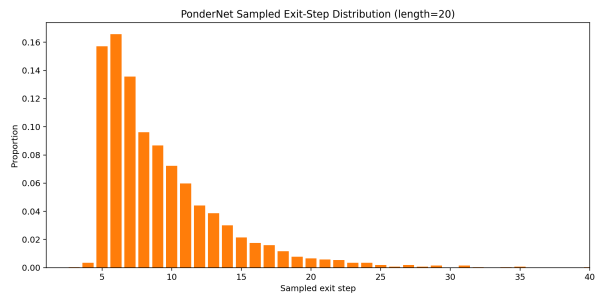
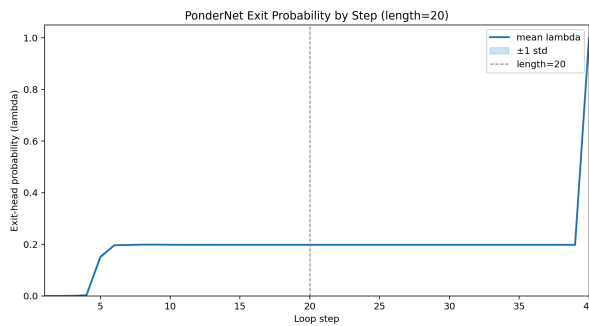
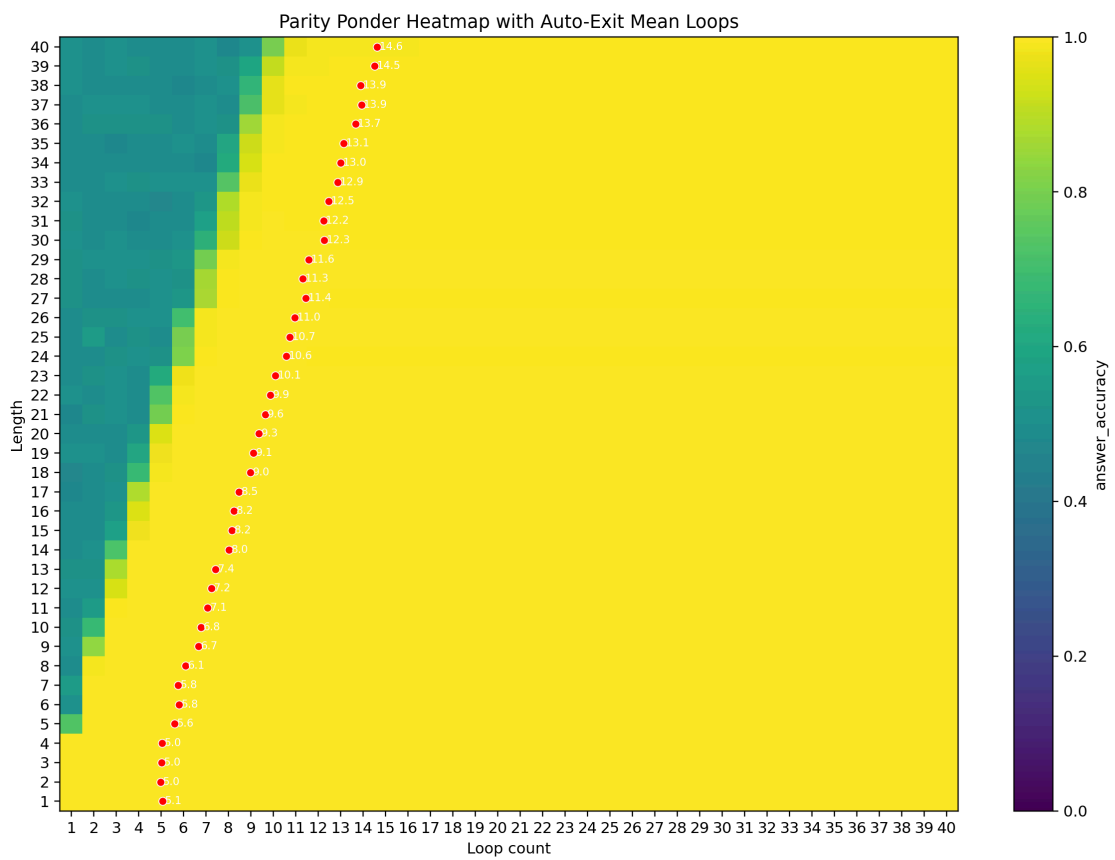
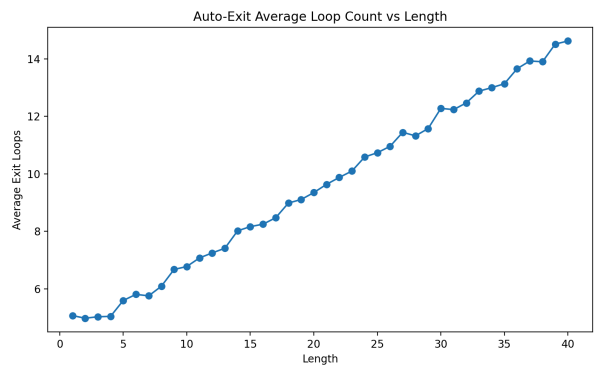
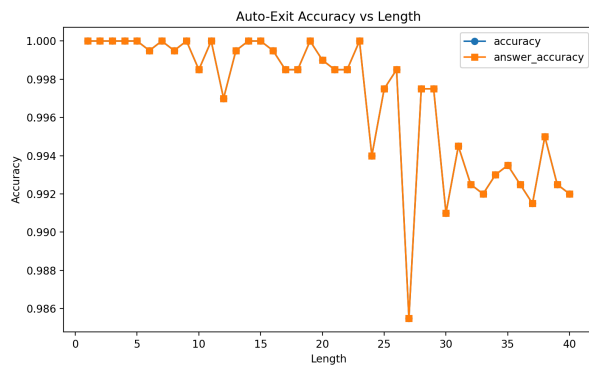
发现他会非常快速的收敛。如果在eval的时候启动自动退出



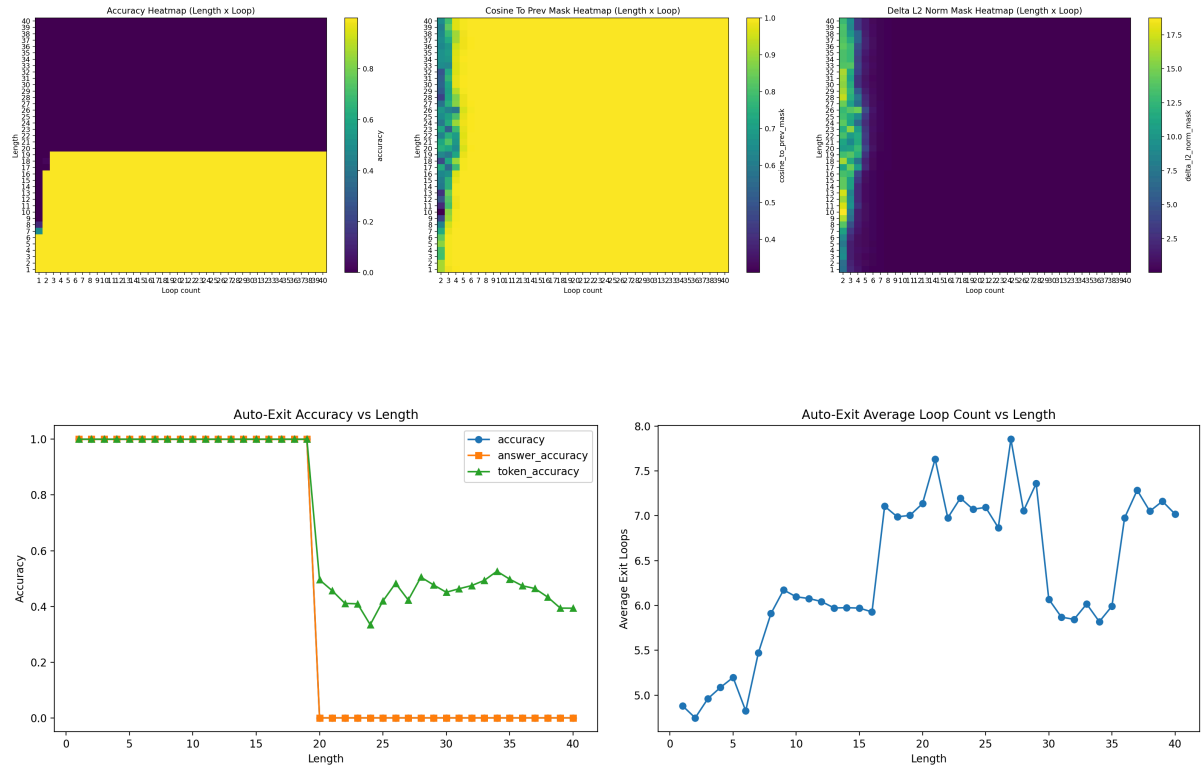
发现随着length的增加，平均loop次数还减少了，而且正确率也很低。

如果是对答案位的第一位接一个Exit head，训练结果是





用PonderNet在Copy任务上尝试了一下



尝试去掉和先验的正则项

